The Lebesgue and Riemann integrals

REAL VARIABLES I

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The Lebesgue integral of a bounded measurable function on $[a, b]$ was built from simple functions, while the Riemann integral was built from lower and upper sums over partitions of $[a, b]$. In these notes we compare the two constructions. We establish Lebesgue's criterion, which characterizes the Riemann integrable functions by the size of their set of discontinuities, and we show that the Riemann and Lebesgue integrals agree whenever the former exists.

The proof expresses the lower Riemann integral of a bounded function as the Lebesgue integral of its lower semicontinuous envelope, and the upper Riemann integral as the Lebesgue integral of its upper semicontinuous envelope. For this purpose we study semicontinuous functions, the envelopes, and an infimal convolution that approximates a lower semicontinuous function from below by Lipschitz functions.

After studying these notes, you will be able to do the following.

- State Lebesgue's criterion for Riemann integrability and use it to decide whether a given bounded function is Riemann integrable.
- Decide whether a function is lower or upper semicontinuous from its level sets.
- Compute the lower and upper semicontinuous envelopes of a bounded function and identify its discontinuity set as the set where they differ.
- Approximate a bounded lower semicontinuous function from below by an increasing sequence of Lipschitz functions using an infimal convolution.
- Prove that the lower and upper Riemann integrals equal the Lebesgue integrals of the lower and upper envelopes.

Throughout, $a < b$ are real numbers, $[a, b]$ carries the Lebesgue measure $\mathcal{L}$, and for $x \in [a, b]$ and $r > 0$ we write

$$B_r(x) \coloneqq (x - r, x + r) \cap [a, b].$$

Open and closed sets in $[a, b]$ are understood relative to the metric $|x - y|$.

10.1 Lebesgue's criterion

For a bounded function $f\colon [a, b] \to \mathbb{R}$ and a partition $P = (a = x_0 < x_1 < \cdots < x_n = b)$, the lower and upper sums are

$$L(f; P) \coloneqq \sum_{j=1}^{n} \Big(\inf_{[x_{j-1}, x_j]} f \Big)(x_j - x_{j-1}), \qquad U(f; P) \coloneqq \sum_{j=1}^{n} \Big(\sup_{[x_{j-1}, x_j]} f \Big)(x_j - x_{j-1}),$$

and the lower and upper Riemann integrals are

$$\underline{I}_R(f) \coloneqq \sup_P L(f; P), \qquad \overline{I}_R(f) \coloneqq \inf_P U(f; P).$$

The function f is Riemann integrable if $\underline{I}_R(f) = \overline{I}_R(f)$, and the common value is its Riemann integral $\int_a^b f(x)\, \mathrm{d}x$. Let

$$D_f \coloneqq \{x \in [a, b] : f \text{ is discontinuous at } x\}$$

denote the set of discontinuities of f.

Theorem 10.1 (Lebesgue's criterion). For a bounded function $f\colon [a,b] \to \mathbb{R}$, the following conditions are equivalent:

1. The function f is Riemann integrable.
2. The set of discontinuities has Lebesgue measure zero: $\mathcal{L}(D_f) = 0$.

Moreover, in this case f is Lebesgue measurable and

$$\int_a^b f(x)\,\mathrm{d}x = \int_{[a,b]} f\,\mathrm{d}\mathcal{L}.$$

The remaining sections develop the tools for the proof, which is given in Section 10.5.

10.2 Semicontinuity

Continuity of g at x requires that for y near x the values $g(y)$ lie above $g(x)-\varepsilon$ and below $g(x)+\varepsilon$. A lower Riemann sum records only the infimum of a function on each subinterval, so it is sensitive to the first of these bounds alone, and an upper sum is sensitive to the second alone. Each bound defines a weaker notion than continuity. For example, $\mathbb{1}_{(0,1)}$ is not continuous at 0, since $\mathbb{1}_{(0,1)}(0) = 0$ while the values at points of $(0,1)$ close to 0 equal 1. The values near 0 nevertheless never fall below $\mathbb{1}_{(0,1)}(0)$, so the lower bound holds at 0 and only the upper bound fails.

Definition 10.2 (Lower and upper semicontinuity). Let $g\colon [a,b] \to \mathbb{R}$ and $x \in [a,b]$. The function g is **lower semicontinuous at** x if for every $\alpha < g(x)$ there exists $r > 0$ such that $g(y) > \alpha$ for all $y \in B_r(x)$. It is **upper semicontinuous at** x if for every $\alpha > g(x)$ there exists $r > 0$ such that $g(y) < \alpha$ for all $y \in B_r(x)$. The function g is **lower semicontinuous** (LSC) or **upper semicontinuous** (USC) if it has the corresponding property at every point.

Lower semicontinuity at x is equivalent to $g(x) \le \liminf_{n\to\infty} g(x_n)$ for every sequence $x_n \to x$. A function is continuous at x if and only if it is both lower and upper semicontinuous at x, since the two conditions together say $g(x)-\varepsilon < g(y) < g(x)+\varepsilon$ near x. Also, g is USC if and only if $-g$ is LSC.

Proposition 10.3 (Level-set characterization). For a function $g\colon [a,b] \to \mathbb{R}$, the following assertions hold:

1. The function g is lower semicontinuous if and only if the superlevel set $\{g > \alpha\}$ is open for every $\alpha \in \mathbb{R}$; equivalently, if and only if the sublevel set $\{g \le \alpha\}$ is closed for every $\alpha \in \mathbb{R}$.
2. The function g is upper semicontinuous if and only if the sublevel set $\{g < \alpha\}$ is open for every $\alpha \in \mathbb{R}$; equivalently, if and only if the superlevel set $\{g \ge \alpha\}$ is closed for every $\alpha \in \mathbb{R}$.

In particular, semicontinuous functions are Borel measurable.

Proof of Proposition 10.3. The two formulations in each item are equivalent because $\{g \le \alpha\}$ is the complement of $\{g > \alpha\}$ and $\{g \ge \alpha\}$ is the complement of $\{g < \alpha\}$. Assume g is LSC and $g(x) > \alpha$. Definition 10.2 gives $r > 0$ with $B_r(x) \subseteq \{g > \alpha\}$, so $\{g > \alpha\}$ is open. Conversely, assume all sets $\{g > \alpha\}$ are open and let $\alpha < g(x)$. Then $x \in \{g > \alpha\}$, so some ball $B_r(x)$ lies in this set, which is lower semicontinuity at x. The statement for USC functions follows by applying this to $-g$. Open sets are Borel, so the sets $\{g > \alpha\}$, respectively $\{-g > \alpha\}$, are Borel for all α. □

Example 10.4 (Indicators). If $O \subseteq [a,b]$ is open, then $\{\mathbb{1}_O > \alpha\}$ is $[a,b]$, O, or $\emptyset$, so $\mathbb{1}_O$ is LSC; equivalently, $\{\mathbb{1}_O \le \alpha\}$ is $\emptyset$, $[a,b]\setminus O$, or $[a,b]$, which are closed. If F is closed, then $\{\mathbb{1}_F \ge \alpha\}$ is $[a,b]$, F, or $\emptyset$, so $\mathbb{1}_F$ is USC. Neither is continuous in general: $\mathbb{1}_{(a,b)}$ is LSC and discontinuous at a and b.

10.3 Semicontinuous envelopes

A bounded function f need not be semicontinuous, but it lies above the constant $\inf f$, which is LSC. We take the largest LSC function below f. This is possible because lower semicontinuity is preserved by suprema.

Lemma 10.5 (Suprema of lower semicontinuous functions). Let $(g_i)_{i\in I}$ be a nonempty family of lower semicontinuous functions on $[a,b]$ whose pointwise supremum $g := \sup_{i\in I} g_i$ is finite. Then g is lower semicontinuous.
Analoguosly, the pointwise infimum of a nonempty family of upper semicontinuous functions is upper semicontinuous whenever it is finite.

Proof of Lemma 10.5. For every $\alpha \in \mathbb{R}$, $g(x) > \alpha$ holds if and only if $g_i(x) > \alpha$ for some i, so $\{g > \alpha\} = \bigcup_{i\in I}\{g_i > \alpha\}$. This is a union of open sets (Proposition 10.3), hence open. The dual statement follows by applying this to $(-g_i)_{i\in I}$. □

Definition 10.6 (Semicontinuous envelopes). Let $f\colon [a,b]\to\mathbb{R}$ be bounded. The **lower semicontinuous envelope** and the **upper semicontinuous envelope** of f are

$$\underline{f}(x) := \sup\{g(x) : g \text{ is LSC},\ g \le f\}, \qquad \overline{f}(x) := \inf\{g(x) : g \text{ is USC},\ g \ge f\}.$$

The family defining $\underline{f}$ contains the constant $\inf f$, and all its members are bounded above by f, so $\underline{f}$ is finite. By Lemma 10.5, $\underline{f}$ is lower semicontinuous. Thus $\underline{f}$ is the greatest LSC function below f, and likewise $\overline{f}$ is the least USC function above f. Since the constants $\inf f$ and $\sup f$ belong to the respective families,

$$\inf f \le \underline{f} \le f \le \overline{f} \le \sup f.$$

By Proposition 10.3, both envelopes are bounded Borel functions. The envelopes are also given by a local formula.

Proposition 10.7 (Local formula for the envelopes). For every bounded $f\colon [a,b]\to\mathbb{R}$ and $x\in[a,b]$,

$$\underline{f}(x) = \sup_{r>0}\ \inf_{y\in B_r(x)} f(y), \qquad \overline{f}(x) = \inf_{r>0}\ \sup_{y\in B_r(x)} f(y).$$

Proof of Proposition 10.7. Let $h(x)$ denote the right-hand side of the first formula. We show that h is an LSC minorant of f and that it dominates every LSC minorant of f. Then $h \le \underline{f}$ by Definition 10.6, and $\underline{f} \le h$ by domination.

Step 1: h is lower semicontinuous and $h \le f$.
Since $x \in B_r(x)$, $\inf_{B_r(x)} f \le f(x)$ for every r, hence $h \le f$. Let $h(x) > \alpha$. By definition of the supremum there is $r > 0$ with $\inf_{B_r(x)} f > \alpha$. For $z \in B_{r/2}(x)$ the triangle inequality gives $B_{r/2}(z) \subseteq B_r(x)$, so

$$h(z) \ge \inf_{B_{r/2}(z)} f \ge \inf_{B_r(x)} f > \alpha.$$

Hence $\{h > \alpha\}$ is open, and h is LSC by Proposition 10.3.

Step 2: h dominates every lower semicontinuous minorant.
Let g be LSC with $g \le f$, and let $\alpha < g(x)$. Choose $r > 0$ with $g > \alpha$ on $B_r(x)$. Then $h(x) \ge \inf_{B_r(x)} f \ge$

$\inf_{B_r(x)} g \geq \alpha$. Since $\alpha < g(x)$ was arbitrary, $g(x) \leq h(x)$.

Step 3: the upper envelope.
A function g is a USC majorant of f if and only if $-g$ is an LSC minorant of $-f$, so $\overline{f} = -\underline{(-f)}$. Applying Steps 1 and 2 to $-f$ and using $\sup(-A) = -\inf A$ gives the second formula. □

The formulas say that $\underline{f}(x) = \liminf_{y\to x} f(y)$ and $\overline{f}(x) = \limsup_{y\to x} f(y)$. The gap between the envelopes therefore measures the oscillation of f at x.

Proposition 10.8 (Discontinuities are the envelope gap). For every bounded $f\colon [a,b] \to \mathbb{R}$, f is continuous at x if and only if $\underline{f}(x) = \overline{f}(x)$. Consequently

$$D_f = \{x \in [a,b] : \underline{f}(x) < \overline{f}(x)\},$$

and D_f is a Borel set.

Proof of Proposition 10.8. Assume f is continuous at x and let $\varepsilon > 0$. Choose $r > 0$ with $|f(y)-f(x)| < \varepsilon$ on $B_r(x)$. By Proposition 10.7,

$$f(x) - \varepsilon \leq \inf_{B_r(x)} f \leq \underline{f}(x) \leq f(x) \leq \overline{f}(x) \leq \sup_{B_r(x)} f \leq f(x) + \varepsilon.$$

Because $\varepsilon > 0$ was arbitrary, $\underline{f}(x) = f(x) = \overline{f}(x)$.
Conversely, assume $\underline{f}(x) = \overline{f}(x)$; then both equal $f(x)$. For $\varepsilon > 0$, Proposition 10.7 gives $r > 0$ with $\inf_{B_r(x)} f > f(x) - \varepsilon$ and $\sup_{B_r(x)} f < f(x) + \varepsilon$ (the smaller of the two radii works for both). Then $|f(y) - f(x)| < \varepsilon$ for $y \in B_r(x)$.
Since $\underline{f} \leq \overline{f}$, the formula for D_f follows. The function $\overline{f} - \underline{f}$ is Borel, so $D_f = \{\overline{f} - \underline{f} > 0\}$ is a Borel set. □

Example 10.9 (Dirichlet's function). Let $f = \mathbb{1}_{\mathbb{Q}\cap[0,1]}$ on $[0,1]$. Every ball $B_r(x)$ contains rational and irrational points, so $\inf_{B_r(x)} f = 0$ and $\sup_{B_r(x)} f = 1$. Hence $\underline{f} = 0$ and $\overline{f} = 1$ everywhere, and $D_f = [0,1]$ by Proposition 10.8.

10.4 Lipschitz approximation from below by infimal convolution

To compute the integral of $\underline{f}$ from Riemann sums, we approximate it from below by functions that are continuous in a quantitative sense. A function g is L-**Lipschitz** if $|g(x) - g(z)| \leq L|x - z|$ for all $x, z \in [a,b]$.

Definition 10.10 (Infimal convolution). For a bounded $f\colon [a,b] \to \mathbb{R}$ and a kernel $k\colon \mathbb{R} \to [0,+\infty]$, the **infimal convolution** of f with k is

$$(\Lambda_k f)(x) \coloneqq \inf_{y\in[a,b]} (f(y) + k(x-y)), \qquad x \in [a,b],$$

with the convention $f(y) + \infty = +\infty$.

The kernel prices the move from x to the competitor point y. We use the kernel $k_n(t) \coloneqq n|t|$, and write $f_n \coloneqq \Lambda_{k_n} f$.

Theorem 10.11 (Lipschitz approximation from below). Let $f\colon [a,b] \to \mathbb{R}$ be bounded and lower semicontinuous, and let

$$f_n(x) \coloneqq \inf_{y\in[a,b]} (f(y) + n|x-y|), \qquad n \in \mathbb{N}.$$

Then each f_n is n-Lipschitz, $\inf f \le f_n \le f_{n+1} \le f$, and $f_n \to f$ pointwise on $[a,b]$.

Proof of Theorem 10.11. Let $m \coloneqq \inf f$. Since $f(y) + n|x-y| \ge m$, each $f_n(x)$ is a real number and $f_n \ge m$.

Step 1: Lipschitz bound and ordering.
For $x, z, y \in [a,b]$ the triangle inequality gives $f(y) + n|x-y| \le f(y) + n|z-y| + n|x-z|$. Taking the infimum over y yields $f_n(x) \le f_n(z) + n|x-z|$, and exchanging x and z gives $|f_n(x) - f_n(z)| \le n|x-z|$. Choosing $y = x$ in the infimum gives $f_n \le f$, and $n|x-y| \le (n+1)|x-y|$ gives $f_n \le f_{n+1}$.

Step 2: pointwise convergence.
Fix x and $\varepsilon > 0$. By lower semicontinuity there is $\delta > 0$ with $f(y) > f(x) - \varepsilon$ whenever $|x-y| < \delta$. Choose N with $m + N\delta \ge f(x)$. For $n \ge N$ and $y \in [a,b]$:

- if $|x-y| < \delta$, then $f(y) + n|x-y| \ge f(y) > f(x) - \varepsilon$;
- if $|x-y| \ge \delta$, then $f(y) + n|x-y| \ge m + N\delta \ge f(x) > f(x) - \varepsilon$.

Taking the infimum over y gives $f(x) - \varepsilon \le f_n(x) \le f(x)$ for $n \ge N$. Because $\varepsilon > 0$ was arbitrary, $f_n(x) \to f(x)$. □

Applied to $-f$, the theorem gives for a bounded USC function f the Lipschitz approximants $\sup_y(f(y) - n|x-y|) \downarrow f$ from above.

Remark (Comparison with the kernel χ_r). Let $\chi_r(t) \coloneqq 0$ for $|t| < r$ and $\chi_r(t) \coloneqq +\infty$ for $|t| \ge r$; thus $\chi_r = +\infty \cdot \mathbb{1}_{\{|t|\ge r\}}$, a hard cutoff in place of the linear penalty. Then

$$(\Lambda_{\chi_r} f)(x) = \inf_{y\in B_r(x)} f(y),$$

so by Proposition 10.7, $\underline{f} = \sup_{r>0} \Lambda_{\chi_r} f$, and $\Lambda_{\chi_{2^{-n}}} f \uparrow \underline{f}$ as $n \to \infty$ because the balls shrink. Both families increase to $\underline{f}$, but the regularity differs. For $f = \mathbb{1}_{(0,1]}$ on $[-1,1]$ and $0 < r < 1/2$, $\Lambda_{\chi_r} f$ equals 0 on $[-1, r)$ and 1 on $[r, 1]$, which is discontinuous, whereas $\Lambda_{k_n} f(x) = \min\{1, \max\{0, nx\}\}$ is n-Lipschitz. The kernel k_n pays a linear price for moving, and this price transfers its Lipschitz constant to f_n.

10.5 Proof of Lebesgue's criterion

The proof compares the lower sums of f with the integral of $\underline{f}$. The first inequality uses that each lower sum is the integral of a step function lying below f that can be taken lower semicontinuous. The second uses the Lipschitz approximants of Theorem 10.11.

Theorem 10.12 (Riemann integrals as integrals of the envelopes). For every bounded $f\colon [a,b] \to \mathbb{R}$,

$$\underline{I}_R(f) = \int_{[a,b]} \underline{f}\, \mathrm{d}\mathcal{L}, \qquad \overline{I}_R(f) = \int_{[a,b]} \overline{f}\, \mathrm{d}\mathcal{L}.$$

Proof of Theorem 10.12. Step 1: $\underline{I}_R(f) \le \int \underline{f}\,\mathrm{d}\mathcal{L}$.
Fix a partition $P = (a = x_0 < \cdots < x_n = b)$ and set $m_j \coloneqq \inf_{[x_{j-1},x_j]} f$. Define $\ell\colon [a,b] \to \mathbb{R}$ by

$$\ell(x) \coloneqq m_j \ \text{ for } x \in (x_{j-1}, x_j), \qquad \ell(x_j) \coloneqq \min\{m_j, m_{j+1}\}\ (0 < j < n), \qquad \ell(x_0) \coloneqq m_1,\ \ell(x_n) \coloneqq m_n.$$

We claim that ℓ is lower semicontinuous and $\ell \le f$. Every point x_j lies in the closed intervals adjacent to it, so $\ell(x_j) \le f(x_j)$, and $\ell \le m_j \le f$ on (x_{j-1}, x_j). For $\alpha \in \mathbb{R}$, the set $\{\ell > \alpha\}$ is the union of the open intervals (x_{j-1}, x_j) with $m_j > \alpha$ together with those partition points x_j whose adjacent values all exceed α. Each such point is an interior point of the union, because the adjacent intervals are included (relative to $[a,b]$ at a and b). Hence $\{\ell > \alpha\}$ is open, and ℓ is LSC by Proposition 10.3.
By Definition 10.6, $\ell \le \underline{f}$. Since ℓ differs from $\sum_j m_j \mathbb{1}_{(x_{j-1},x_j)}$ only on a finite set,

$$L(f;P) = \int_{[a,b]} \ell\,\mathrm{d}\mathcal{L} \le \int_{[a,b]} \underline{f}\,\mathrm{d}\mathcal{L}.$$

Taking the supremum over P gives the claim.

Step 2: $\int \underline{f}\,\mathrm{d}\mathcal{L} \le \underline{I}_R(f)$.
By Theorem 10.11 applied to the bounded LSC function $\underline{f}$, there are n-Lipschitz functions f_n with $\inf \underline{f} \le f_n \le \underline{f} \le f$ and $f_n \uparrow \underline{f}$ pointwise. Fix n and a partition P into k intervals of length $h = (b-a)/k$. For every interval $I_j = [x_{j-1}, x_j]$ and $x, z \in I_j$, the Lipschitz bound gives $f_n(x) \le f_n(z) + nh$, hence $f_n(x) \le \inf_{I_j} f_n + nh$ on I_j. Integrating over I_j and summing,

$$\int_{[a,b]} f_n\,\mathrm{d}\mathcal{L} \le L(f_n;P) + nh(b-a) \le L(f;P) + nh(b-a) \le \underline{I}_R(f) + nh(b-a),$$

where $L(f_n;P) \le L(f;P)$ because $f_n \le f$. Letting $k \to \infty$ gives $\int f_n\,\mathrm{d}\mathcal{L} \le \underline{I}_R(f)$. The functions $f_n - \inf \underline{f}$ are non-negative, continuous, and increase to $\underline{f} - \inf \underline{f}$, so the monotone convergence theorem yields

$$\int_{[a,b]} \underline{f}\,\mathrm{d}\mathcal{L} = \lim_{n\to\infty} \int_{[a,b]} f_n\,\mathrm{d}\mathcal{L} \le \underline{I}_R(f).$$

Steps 1 and 2 prove the first identity.

Step 3: the upper integral.
For every partition P, $L(-f;P) = -U(f;P)$, so $\underline{I}_R(-f) = -\overline{I}_R(f)$. Also $\underline{(-f)} = -\overline{f}$ by Proposition 10.7. The first identity applied to $-f$ gives $-\overline{I}_R(f) = -\int \overline{f}\,\mathrm{d}\mathcal{L}$. □

Proof of Theorem 10.1. Step 1: equivalence.
By Theorem 10.12, f is Riemann integrable if and only if $\int \underline{f}\,\mathrm{d}\mathcal{L} = \int \overline{f}\,\mathrm{d}\mathcal{L}$, that is, $\int (\overline{f} - \underline{f})\,\mathrm{d}\mathcal{L} = 0$. The function $\overline{f} - \underline{f}$ is non-negative and Borel, so its integral vanishes if and only if it vanishes almost everywhere (Proposition 8.6). By Proposition 10.8, $\{\overline{f} - \underline{f} > 0\} = D_f$, so f is Riemann integrable if and only if $\mathcal{L}(D_f) = 0$.

Step 2: agreement of the integrals.
Assume $\mathcal{L}(D_f) = 0$. Since $\underline{f} \le f \le \overline{f}$ and $\underline{f} = \overline{f}$ outside D_f, we have $f = \underline{f}$ almost everywhere. As $\underline{f}$ is Borel and Lebesgue measure is complete, f is Lebesgue measurable, and functions that agree almost everywhere have the same integral. By Theorem 10.12,

$$\int_{[a,b]} f\,\mathrm{d}\mathcal{L} = \int_{[a,b]} \underline{f}\,\mathrm{d}\mathcal{L} = \underline{I}_R(f) = \int_a^b f(x)\,\mathrm{d}x.$$ □

For the Dirichlet function of Example 10.9, Theorem 10.12 gives $\underline{I}_R(f) = 0$ and $\overline{I}_R(f) = 1$, so f is not Riemann integrable, although it is Lebesgue integrable with integral 0 because $\mathbb{Q} \cap [0,1]$ is null. Boundedness cannot be dropped from the criterion: $x^{-1/2}$ on $(0,1]$, with value 0 at $x = 0$, is discontinuous only at 0, yet it is unbounded and therefore has no proper Riemann integral, while its Lebesgue integral equals 2.